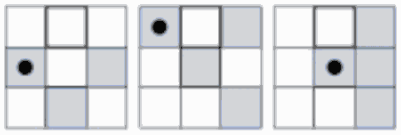
9.1 Find the reflection, $\hat{B}$, of each of the following structuring elements. The dot indicates the origin of the SE.



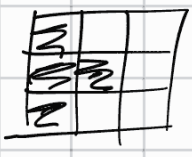
(a)* (b) (c)

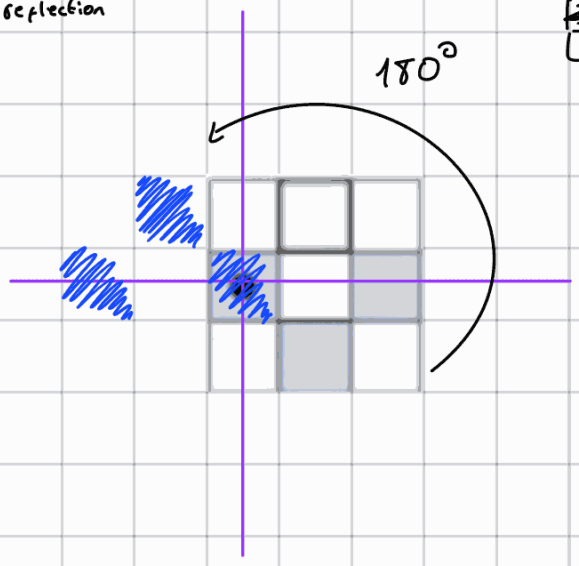
$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$

reflection

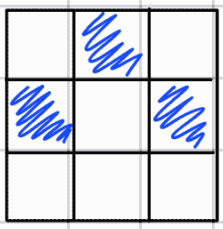


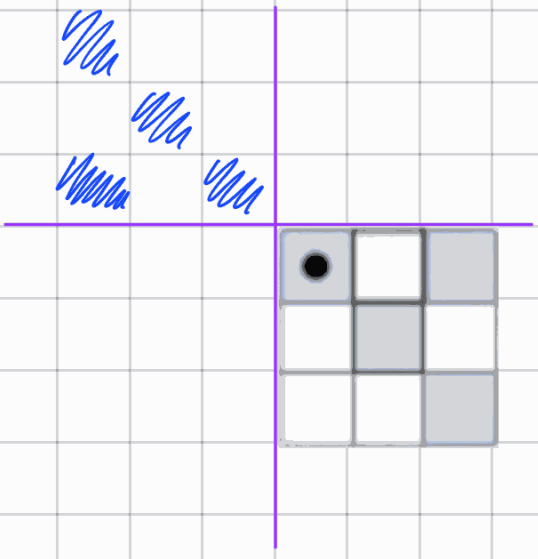


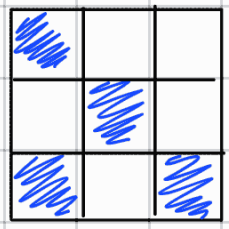


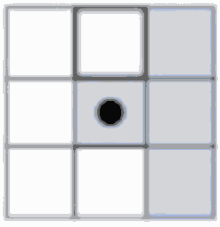


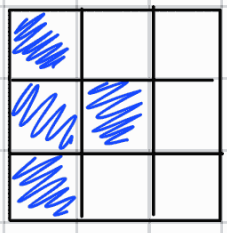
⇒











9.32 Answer the following:

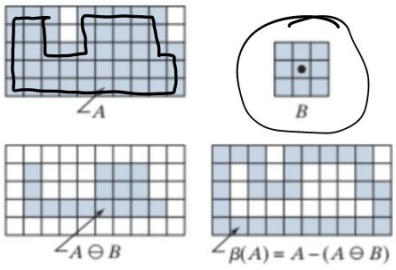
(a)* Discuss the effect of using the structuring element in Fig. 9.17(c) for boundary extraction, instead of the element in Fig. 9.15(b).

(b) What would be the effect of using a 3×3 structuring element composed of all 1's in the hole filling algorithm of Eq. (9-19), instead of the structuring element in Fig. 9.17(c)?

$$X_k = (X_{k-1} \oplus B) \cap I^c \qquad k = 1, 2, 3, \ldots \qquad (9\text{-}19)$$

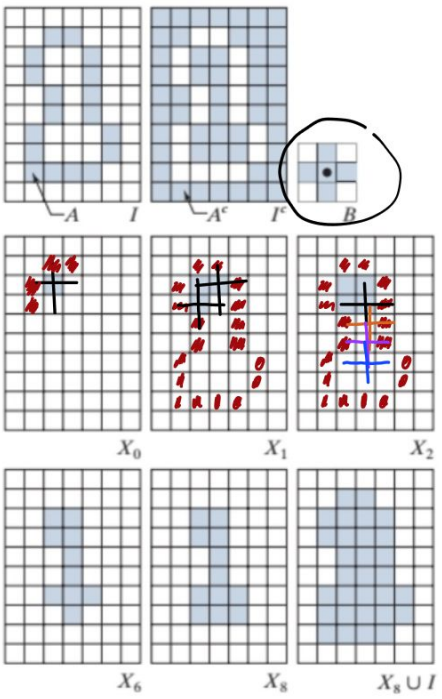
a b
c d

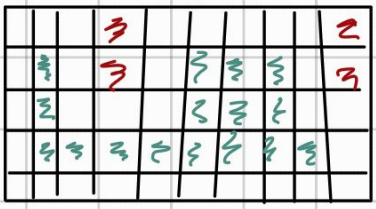
FIGURE 9.15
(a) Set, A, of foreground pixels.
(b) Structuring element.
(c) A eroded by B.
(d) Boundary of A.

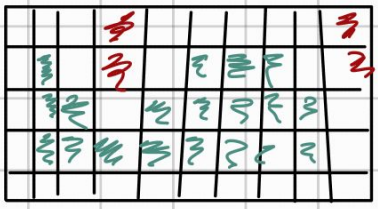


a b c
d e f
g h i

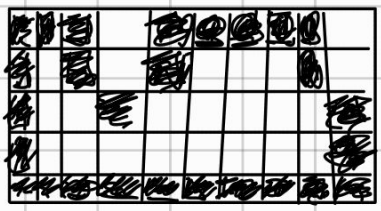
FIGURE 9.17
Hole filling.
(a) Set A (shown shaded) contained in image I.
(b) Complement of I.
(c) Structuring element B. *Only* the foreground elements are used in computations
(d) Initial point inside hole, set to 1.
(e)–(h) Various steps of Eq. (9-19).
(i) Final result [union of (a) and (h)].



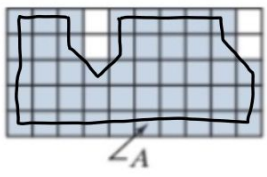


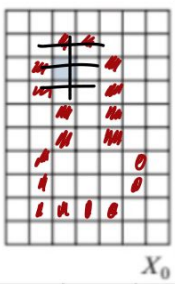


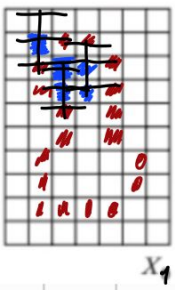
A ⊖ B

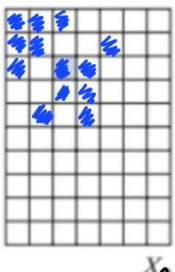


A - (A⊖B)









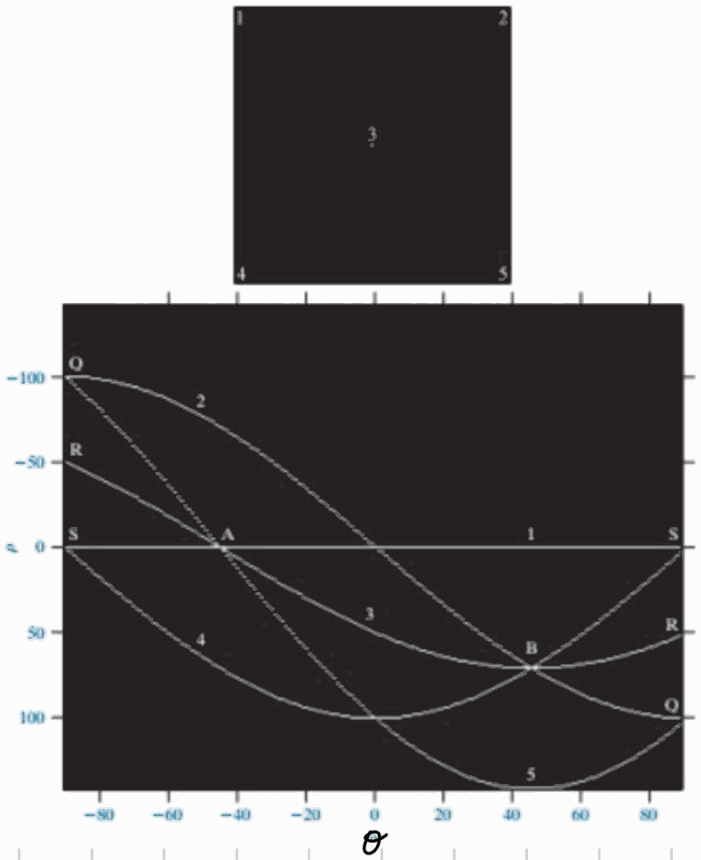
bu aşamada yapı elemanının gapısı yüzden taşma gerçekleşmektedir ve başarı ile boşluk doldurulamacaktır son aşamada oluşacak görüntü A^c ye eşit olacaktır.

10.26 Refer to the Hough transform in Section 10.2.

(a)* Explain why the Hough mapping of the point labeled 1 in Fig. 10.30(a) is a straight line in Fig. 10.30(b).

(b)* Is this the only point that would produce that result? Explain ✓ Yes it will

(c) Explain the reflective adjacency relationship illustrated by, for example, the curve labeled Q in Fig. 10.30(b).



a
b

FIGURE 10.30
(a) Image of size 101 × 101 pixels, containing five white points (four in the corners and one in the center). (b) Corresponding parameter space.

$$\rho = x\cos\theta + y\sin\theta$$

1 noktasına azaklık 0 tüm açılardan olduğundan $\rho = 0$ 'da bir çizgi olarak gözükmektedir

Because it is located in the origin thats why it's distance is zero. And also its' angle is zero too. Think of it like where ever you turn from that point it will always be zero

+

$$\rho = x\cos\theta + y\sin\theta$$

(1) $\Rightarrow \rho = 0$

(2) $\Rightarrow \rho = 0\cos\theta + 100\sin\theta$

$$\rho_2 = 100\sin\theta$$

(5) $\Rightarrow \rho_5 = 100\cos\theta + 100\sin\theta$

$$\rho_2 = \rho_5 \Rightarrow 100\sin\theta = \underbrace{100\cos\theta}_{0^{\circ}} + 100\sin\theta$$